Module - 5
Deep learning.

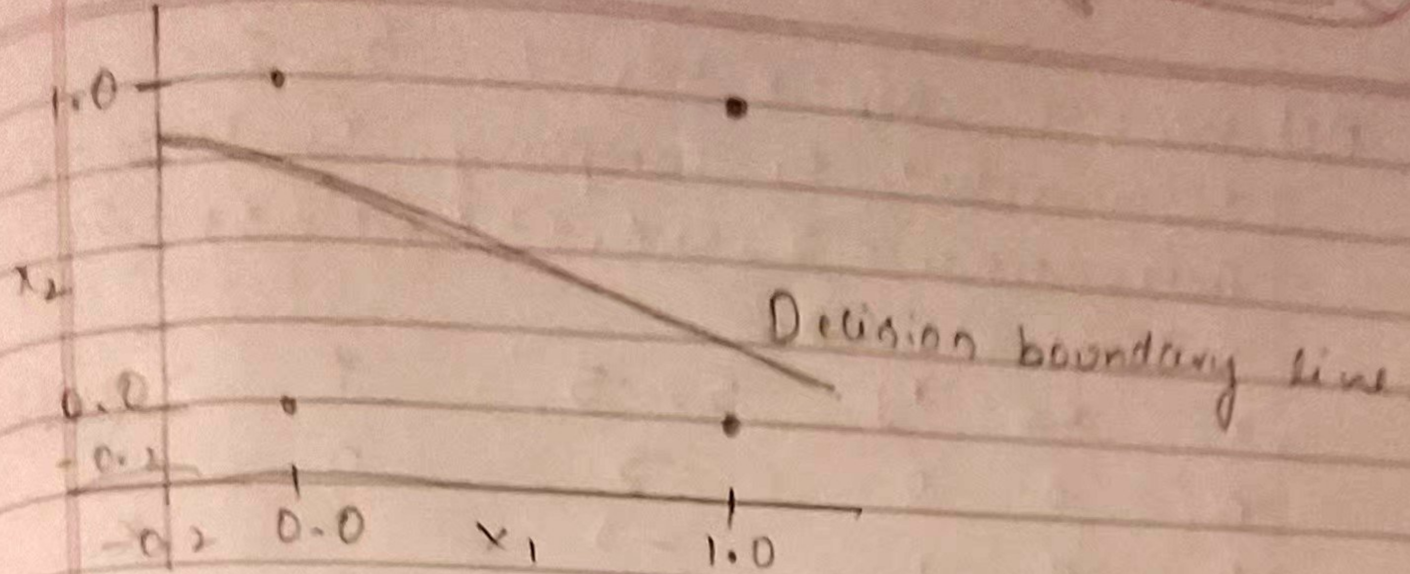
P1) $w_1 = 4$, $w_2 = 2$, $w_3 = 2$, $b = -5$,
x_1 = weather, x_2 = Company, x_3 = Proximity
Activation = $S = 4x_1 + 2x_2 + 2x_3 - 5$
O.P = 1 if $S > 0$.

x_1	x_2	x_3	$S = -5$.	Y	result	
0	0	0	-5	0	x	
0	0	1	-3	0	x	
0	1	0	-3	0	x	
0	1	1	-1	0	x	
1	0	0	-1	0	x	
1	0	1	1	1	✓	Go
1	1	0	1	1	✓	Go
1	1	1	3	1	✓	Go.

→ (1.0.1), (1.1.0) (1.1.1).

P2] Inputs = x_1 & x_2. weights = $w_1 = 1$, $w_2 = 2$
Bias. $b = -1.5$.
→ $S = w_1x_1 + w_2x_2 + b = x_1 + 2x_2 - 1.5$
$y = 1$ if $S > 0$.
→ Decision boundary Equation.
$x_1 + 2x_2 - 1.5 = 0 \Rightarrow x_2 = \frac{1.5 - x_1}{2}$

x_1	x_2	$b = -1.5$.	y
0	0	-1.5	0
0	1	0.5	1
1	0	-0.5	0
1	1	1.5	1



P3] logic Gate Emulation.

$w_1 = w_2 = -2$ $b = 3$ in NAND Gate

NAND = NOT + AND Gate

$S = w_1 x_1 + w_2 x_2 + b = -2x_1 - 2x_2 + 3$

$y = 1$ if $S > 0$ else 0

→ Truth table.

x_1	x_2	$S = -2x_1 - 2x_2 + 3$	Y
0	0	3	1
0	1	1	1
1	0	1	1
1	1	-1	0

→ others Gates are :- AND Gate, NOT Gate, OR Gate.

P4] Non-linearly separable Patterns

XOR - function

→ XOR Truth Table.

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0

$y = 1$ when inputs are different

$$y = \begin{cases} 1 & S > 0 \quad w_1x_1 + w_2x_2 + b > 0 \\ 0 & \end{cases}$$

→ XOR is non-linearly separable - The 1's & 0's can't be separated by single straight line in 2D.

We need multi-layer perceptron to implement XOR.

- $w_1x_1 + w_2x_2 + b$

→ i) $w_2 + b > 0$ ii) $w_1 + b < 0$. iii) $b \leq 0$.

iv) $w_1 + w_2 + b \leq 0$.

$w_1 + b > 0 \Rightarrow b > -w_1$, $w_2 + b > 0 \Rightarrow b > -w_2$

$b \leq 0$.

$(w_1 + b) + (w_2 + b) \leq -b$

LHS '+' RHS '−'

Contradiction.

No solution.